

Periodic function or fundamental period

A function $f(x)$ is said to be periodic if and only if $f(x+p) = f(x)$ is true for some value of p and every value of x .

The smallest value of p for which this equation is true for every values of x this is called periodic of the function $f(x)$

Dirichlet's condition.

(i) $f(x)$ is periodic, single valued and finite

(ii) $f(x)$ has a finite no. of discontinuity in anyone period and no infinite discontinuity

(iii) $f(x)$ has atmost a finite no. of maximum or minimum

$$\text{Type I} \begin{cases} (0, 2\pi) \\ (-\pi, \pi) \end{cases}$$

$$\text{Change of Interval} \leftarrow \text{Type II} \begin{cases} (0, 2l) \\ (-l, l) \end{cases}$$

$$\text{Half Range Series} \leftarrow \text{Type III} \begin{cases} (0, \pi) \\ (0, l) \end{cases}$$

Fourier series for $(0, 2\pi)$

The Fourier series for the fun $f(x)$ in the interval $(0, 2\pi)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$\text{where } a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

Deduction :

$(0, 2\pi)$

$x=0$ is an discontinuity pt

$\therefore$ Fourier series at $x=0$ converges to $\frac{f(0)+f(2\pi)}{2}$

$(-\pi, \pi)$

(i) $x=0$ is continuity pt

$\therefore$ Fourier Series at $x=0$ converges to $f(0)$.

(ii) $x=\pi$ is discontinuity pt

$\therefore$ F.S at $x=\pi$ converges to $\frac{f(-\pi) + f(\pi)}{2}$

Problem:

$f(x) = (\pi - x)^2$ in $(0, 2\pi)$. deduce the sum of series $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

Solu

w.k.T

Fourier series for the fun $f(x)$ in the interval $(0, 2\pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \quad \rightarrow \textcircled{1}$$

find a_0 :

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 dx$$

$$= \frac{1}{\pi} \left[\frac{(\pi - x)^3}{-3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[-\frac{1}{3} (\pi - x)^3 \right]_0^{2\pi}$$

$$= \frac{-1}{3\pi} \left[(\pi - 2\pi)^3 - (\pi - 0)^3 \right]$$

$$= \frac{-1}{3\pi} \left[(-\pi)^3 - (\pi)^3 \right]$$

$$= \frac{-1}{3\pi} (-\pi^3 - \pi^3)$$

$$= \frac{-1}{3\pi} (-2\pi^3)$$

$$\boxed{a_0 = \frac{2\pi^2}{3}}$$

To find a_n

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 \cos nx \, dx$$

By parts then

$$u = (\pi - x)^2$$

$$u' = -2(\pi - x)$$

$$u'' = -2(-1) = 2$$

$$dv = \cos nx$$

$$v = \frac{\sin nx}{n}$$

$$v_1 = \frac{-\cos nx}{n^2}$$

$$v_2 = \frac{-\sin nx}{n^3}$$

W.K.T

$$\int u \, dv = uv - u'v_1 + u''v_2 - \dots$$

$$a_n = \frac{1}{\pi} \left[(\pi-x)^2 \frac{\sin nx}{n} - 2(\pi-x) \frac{\cos nx}{n^2} - 2 \frac{\sin nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \frac{1}{n} \left[(\pi-x)^2 \sin nx \right]_0^{2\pi} - \frac{2}{n^2} \left[(\pi-x) \cos nx \right]_0^{2\pi} - \frac{2}{n^3} \left[\sin nx \right]_0^{2\pi} \right\}$$

$$\sin 2\pi = 0$$

$$\sin \pi = 0$$

$$\sin 0 = 0$$

$$\cos 2\pi = 1$$

$$\cos \pi = -1$$

$$\cos 0 = 1$$

$$\cos(\pi+1)\pi = -(-1)$$

$$\cos(\pi-1)\pi = -(-1)$$

$$= \frac{1}{\pi} \left\{ \frac{1}{n} \left[(\pi-2\pi)^2 \sin n(2\pi) - (\pi-0)^2 \sin n(0) \right] \right.$$

$$- \frac{2}{n^2} \left[(\pi-2\pi) \cos n(2\pi) - (\pi-0) \cos n(0) \right]$$

$$- \frac{2}{n^3} \left[\sin n(2\pi) - \sin n(0) \right] \left. \right\}$$

$$= \frac{+1}{\pi} \left\{ \frac{-2}{n^2} \left[(\pi-2\pi) - (\pi-0) \right] \right.$$

$$= \frac{+1}{\pi} \left[\frac{-2}{n^2} (-\pi - \pi) \right]$$

$$= \frac{+1}{\pi} \left[\frac{-2}{n^2} (-2\pi) \right]$$

$$= \frac{+4}{n^2}$$

$$a_n = \frac{4}{n^2}$$

Find b_n

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 \sin nx \, dx$$

$$u = (\pi - x)^2$$

$$dv = \sin nx$$

$$u' = -2(\pi - x)$$

$$v = -\frac{\cos nx}{n}$$

$$u'' = -2(-1) = 2$$

$$v_1 = -\frac{\sin nx}{n^2}$$

$$v_2 = \frac{\cos nx}{n^3}$$

$$\int u dv = uv - u'v_1 + u''v_2$$

$$b_n = \frac{1}{\pi} \left[-(\pi - x)^2 \frac{\cos nx}{n} - 2(\pi - x) \frac{\sin nx}{n^2} + 2 \frac{\cos nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ -\frac{1}{n} \left[(\pi - x)^2 \cos nx \right]_0^{2\pi} - \frac{2}{n^2} \left[(\pi - x) \sin nx \right]_0^{2\pi} + \frac{2}{n^3} \left[\cos nx \right]_0^{2\pi} \right\}$$

$$= \frac{+1}{\pi} \left\{ \frac{-1}{n} \left[(\pi - 2\pi)^2 \cos n(2\pi) - (\pi - 0)^2 \cos n(0) \right] \right.$$

$$\left. - \frac{2}{n^2} \left[(\pi - 2\pi) \sin n(2\pi) - (\pi - 0) \sin n(0) \right] \right\}$$

$$+ \frac{2}{n^3} \left[\underbrace{\cosh 2\lambda}_1 - \underbrace{\cosh(0)}_1 \right] \}$$

$$= \frac{1}{\lambda} \left\{ \frac{-1}{n} [(-\lambda)^2 - (\lambda)^2] + \frac{2}{n^3} (1 - 1) \right\}$$

$$= \frac{1}{\lambda} \left[\frac{-1}{n} (+\cancel{\lambda^2} - \lambda^2) \right]$$

$$= \frac{1}{\lambda} \left[\frac{-1}{n} (0) \right]$$

$$= \cancel{\frac{1}{\lambda}} \cancel{\frac{2\lambda^2}{n}}$$

$$b_n = 0$$

Sub above values in equ (1)

$$f(x) = \frac{1}{2} \left(\frac{2\lambda^2}{3} \right) + \sum_{n=1}^{\infty} \frac{4}{n^2} \cosh nx + \sum_{n=1}^{\infty} (0) \sinh nx$$

$$f(x) = \frac{\lambda^2}{3} + 4 \sum_{n=1}^{\infty} \frac{\cosh nx}{n^2}$$

Deduction:

$$\frac{\lambda^2}{3} + 4 \sum_{n=1}^{\infty} \frac{\cosh nx}{n^2} = f(x)$$

$$\frac{\lambda^2}{3} + 4 \left[\frac{\cosh x}{1^2} + \frac{\cosh 2x}{2^2} + \frac{\cosh 3x}{3^2} + \dots \right] = f(x)$$

$$P_n(0, 2\pi)$$

Fourier series $x=0$ converges to $\frac{f(0) + f(2\pi)}{2}$

$$\frac{\pi^2}{3} + 4 \left[\frac{\cos 0}{1^2} + \frac{\cos 0}{2^2} + \frac{\cos 0}{3^2} + \dots \right] = \frac{f(0) + f(2\pi)}{2}$$

$$f(x) = (\pi - x)^2$$

$$f(0) = (\pi - 0)^2 = \pi^2$$

$$f(2\pi) = (\pi - 2\pi)^2 = (-\pi)^2 = \pi^2$$

$$\frac{\pi^2}{3} + 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = \frac{\pi^2 + \pi^2}{2}$$

$$4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \dots \right] = \frac{2\pi^2}{2} - \frac{\pi^2}{3}$$

$$4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \dots \right] = \frac{3\pi^2 - \pi^2}{3}$$

$$4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \dots \right] = \frac{2\pi^2}{3}$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \dots = \frac{2\pi^2}{3} \times \frac{1}{4}$$

$$= \frac{\pi^2}{6}$$

7) $f(x) = \left(\frac{\pi-x}{2}\right)^2$ in $(0, 2\pi)$. Deduce that $\frac{1}{1^2} + \frac{1}{2^2} + \dots = \frac{\pi^2}{6}$
 w.k.T

8) Fourier series for the fun $f(x)$ in the interval $(0, 2\pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \rightarrow \textcircled{1}$$

So find a_0 :

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} \left(\frac{\pi-x}{2}\right)^2 dx$$

$$= \frac{1}{\pi} \cdot \frac{1}{4} \int_0^{2\pi} (\pi-x)^2 dx$$

$$= \frac{1}{4\pi} \left[\frac{(\pi-x)^3}{-3} \right]_0^{2\pi}$$

$$= \frac{-1}{12\pi} \left[(\pi-x)^3 \right]_0^{2\pi}$$

$$= \frac{-1}{12\pi} \left[(\pi-2\pi)^3 - (\pi-0)^3 \right]$$

$$= \frac{-1}{12\pi} \left[(-\pi)^3 - (\pi)^3 \right]$$

$$= \frac{-1}{12\pi} \left[-\pi^3 - \pi^3 \right]$$

$$= \frac{-1}{\frac{12\pi}{6}} (-2\pi^2)$$

$$a_0 = \pi^2/6$$

To find a_n

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} \left(\frac{\pi-x}{2}\right)^2 \cos nx \, dx$$

$$= \frac{1}{4\pi} \int_0^{2\pi} (\pi-x)^2 \cos nx \, dx$$

$$u = (\pi-x)^2$$

$$dv = \cos nx$$

$$u' = -2(\pi-x)$$

$$v = \frac{\sin nx}{n}$$

$$u'' = -2(-1) = 2$$

$$V_1 = \frac{-\cos nx}{n^2}$$

$$V_2 = \frac{-\sin nx}{n^3}$$

$$a_n = \frac{1}{4\pi} \left[(\pi-x)^2 \frac{\sin nx}{n} - 2(\pi-x) \frac{\cos nx}{n^2} - 2 \frac{\sin nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{4\pi} \left\{ \frac{1}{n} \left[(\pi-x)^2 \sin nx \right]_0^{2\pi} - \frac{2}{n^2} \left[(\pi-x) \cos nx \right]_0^{2\pi} - \frac{2}{n^3} \left[\sin nx \right]_0^{2\pi} \right\}$$

$$= \frac{1}{4\pi} \left\{ \frac{1}{n} \left[\underbrace{(\pi-2\pi)^2}_{0} \sin \underbrace{\pi}_{0} - (\pi-0)^2 \sin \underbrace{\pi(0)}_{0} \right] \right. \\ \left. - \frac{2}{n^2} \left[(\pi-2\pi) \underbrace{\cosh 2\pi}_1 - (\pi-0) \underbrace{\cosh(0)}_1 \right] \right. \\ \left. - \frac{2}{n^3} \left[\underbrace{\sin \pi}_{0} - \underbrace{\sin \pi(0)}_{0} \right] \right\}$$

$$= \frac{1}{4\pi} \left\{ \frac{-2}{n^2} (-\pi - \pi) \right\}$$

$$= \frac{1}{4\pi} \left[\frac{-2}{n^2} (-2\pi) \right]$$

$$a_n = \frac{1}{n^2}$$

to find b_n

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} \left(\frac{\pi-x}{2} \right)^2 \sin nx \, dx$$

$$= \frac{1}{4\pi} \int_0^{2\pi} (\pi-x)^2 \sin nx \, dx$$

$$u = (\pi-x)^2 \quad \begin{array}{l} dv = \sin nx \\ v = \frac{-\cos nx}{n} \end{array}$$

$$u' = -2(\pi-x) \quad v_1 = \frac{-\sin nx}{n^2}$$

$$u'' = -2(-1) = 2 \quad v_2 = \frac{\cos nx}{n^3}$$

$$= \frac{1}{4\pi} \left\{ -\frac{(\pi-x)^2 \cosh nx}{n} - 2(\pi-x) \frac{\sinh nx}{n^2} + 2 \frac{\cosh nx}{n^3} \right\}$$

$$= \frac{1}{4\pi} \left\{ -\frac{1}{n} \left[(\pi-x)^2 \cosh nx \right]_0^{\pi} - \frac{2}{n^2} \left[(\pi-x) \sinh nx \right]_0^{\pi} + \frac{2}{n^3} \left[\cosh nx \right]_0^{\pi} \right\}$$

$$= \frac{1}{4\pi} \left\{ \frac{-1}{n} \left[(\pi-2\pi)^2 \cosh 2\pi - (\pi-0)^2 \cosh(0) \right] - \frac{2}{n^2} \left[(\pi-2\pi) \sinh 2\pi - (\pi-0) \sinh(0) \right] + \frac{2}{n^3} \left[\cosh 2\pi - \cosh(0) \right] \right\}$$

$$= \frac{1}{4\pi} \left\{ \frac{-1}{n} \left[(-\pi)^2 - (\pi)^2 \right] + \frac{2}{n^3} (1-1) \right\}$$

$$= \frac{1}{4\pi} \left[\frac{-1}{n} (+\pi^2 - \pi^2) \right]$$

$$= \frac{1}{4\pi} \left[\frac{-1}{n} (0) \right]$$

$$b_n = 0$$

Sub above value in equ (1)

$$f(x) = \frac{1}{2} \left(\frac{\pi^2}{6} \right) + \sum_{n=1}^{\infty} \frac{1}{n^2} \cosh nx + \sum_{n=1}^{\infty} (0) \sinh nx$$

$$f(x) = \frac{1}{2} \left(\frac{\pi^2}{6} \right) + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$$

reduction:

$$\frac{\pi^2}{12} + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2} = f(x)$$

$$\frac{\pi^2}{12} + \left[\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} + \dots \right] = f(x)$$

in $(0, 2\pi)$

Fourier series $x=0$ converges to $\frac{f(0) + f(2\pi)}{2}$

$$\frac{\pi^2}{12} + \left[\frac{\cos 0}{1^2} + \frac{\cos 0}{2^2} + \frac{\cos 0}{3^2} + \dots \right] = \frac{f(0) + f(2\pi)}{2}$$

$$f(x) = \left(\frac{\pi - x}{2} \right)^2$$

$$f(0) = \left(\frac{\pi - 0}{2} \right)^2 = \frac{\pi^2}{4}$$

$$f(2\pi) = \left(\frac{\pi - 2\pi}{2} \right)^2 = \left(-\frac{\pi}{2} \right)^2 = \frac{\pi^2}{4}$$

$$\frac{\pi^2}{12} + \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = \frac{\pi^2/4 + \pi^2/4}{2}$$

$$\frac{\pi^2}{12} + \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = 2 \frac{\pi^2}{8}$$

$$\left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] = \frac{1}{8} \pi^2 - \frac{\pi^2}{12}$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{4} - \frac{\pi^2}{12}$$

$$= \frac{3\pi^2 - \pi^2}{12}$$

$$= \frac{2\pi^2}{12}$$

$$= \pi^2/6$$

2. $f(x) = x(2\pi - x)$ in $(0, 2\pi)$ Deduce that

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \pi^2/6$$

Solu.

w.k.T

Fourier series for the fun $f(x)$ in the interval $(0, 2\pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx \rightarrow (1)$$

To find a_0

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} x(2\pi - x) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} (2\pi x - x^2) dx$$

$$= \frac{1}{\pi} \left(2\pi \frac{x^2}{2} - \frac{x^3}{3} \right)_0^{2\pi}$$